NIKUNJ GARG

Java Backend Developer | BCA Student

Dehradun, India • +91 94565 00319 • gargnikunj991@gmail.com • Portfolio: nikunjgarg.xyz • LeetCode: @Nikunjgarg12 • GitHub • LinkedIn

SUMMARY

BCA student with hands-on experience developing high-concurrency backend systems using **Java 21**, **Spring Boot 3**, **PostgreSQL**, and **Redis**. Built systems involving **distributed locks (Redisson)**, ACID concurrency control, idempotency gates, stateless **JWT** security, and real-time **WebSocket (STOMP)** messaging. Backed by automated **JUnit 5** stress and failure-mode testing.

TECHNICAL SKILLS

- **Languages:** Java 21, SQL
- **Backend & Distributed:** Spring Boot 3, Spring Data JPA, Hibernate ORM, RESTful APIs, WebSockets (STOMP), Distributed Locks (Redisson)
- **Databases & Caching:** PostgreSQL, Redis 7 (TTL Leases, Keyspace Events), Schema Design (3NF), ACID Transactions, Locking (Pessimistic/Optimistic)
- **Security & Testing:** Spring Security, JWT (Refresh Token Rotation), Role-Based Access Control (RBAC), JUnit 5 (70 Tests), Mockito
- **Tools & DevOps:** Docker, Docker Compose, Git, GitHub, Maven, Postman, Micrometer, OpenAPI 3.0, Render

PROJECTS

DevTinder — Developer Matching & Messaging Platform | Java 21, Spring Boot 3, PostgreSQL, WebSockets (STOMP), Spring Security, Docker GitHub Repo ↗

- Built bidirectional real-time messaging using **Spring WebSocket (STOMP)** paired with REST delta synchronization (`?after={id}`) and PostgreSQL persistence so users can retrieve missed messages upon reconnecting.
- Implemented a developer discovery and matching service that ranks candidate profiles dynamically by shared skill counts in descending order and tracks mutual interest using unique composite constraints on swipe records.
- Hardened authentication using stateless **Spring Security** filters, short-lived **JWT** access tokens, sliding-window IP rate limiting, and **Refresh Token Rotation (RTR)** with token family tracking to detect and revoke compromised sessions.
- Containerized the backend using a multi-stage **Dockerfile** and configured environment variables and health-check endpoints (`/api/health`) for cloud deployment integrating with a Next.js frontend.

LibroSphere — Library Management & Reservation System | Java 21, Spring Boot 3, PostgreSQL, Spring Data JPA, Spring Security, Docker, JUnit 5 GitHub Repo ↗

- Implemented pessimistic row-level locking (`SELECT ... FOR UPDATE`) in **Spring Data JPA** within atomic `@Transactional` boundaries to prevent race conditions and inventory overselling during concurrent checkouts.
- Verified thread safety and ACID resilience with an automated **JUnit 5** integration test using `CountDownLatch` and `ExecutorService`, proving 10 concurrent threads competing for 1 copy produced exactly 1 loan and 9 rejections with zero underflow.
- Architected an automated FIFO reservation waitlist that intercepts returned books, holding them exclusively for queued patrons with a 48-hour pickup window before restoring unclaimed inventory to general circulation.
- Built an automated nightly cron worker (`@Scheduled`) calculating tiered overdue penalties and added composite B-Tree indexes on `(returned, due_date)` and `(book_id, status, reserved_at)` to avoid full table scans.

TicketWave — Distributed Seat Hold & Checkout Engine | Java 21, Spring Boot 3, Redis 7 (Redisson), PostgreSQL, Micrometer, JUnit 5 GitHub Repo ↗

- Architected distributed seat hold allocation using **Redisson MultiLock** across Redis (`<1ms`) with deterministic seat-ID sorting to eliminate deadlocks under high-concurrency arena ticket spikes.
- Built a **Redis SETNX idempotency gate** (`IN_PROGRESS` 120s / `COMPLETED` 24h) and decoupled payment execution outside DB boundaries, preventing HikariCP pool exhaustion with atomic PostgreSQL commits.
- Engineered a **dual-tier inventory reclamation** pipeline (Redis keyspace pub/sub + scheduled PostgreSQL sweeper) to auto-reclaim expired 10m holds; verified zero overselling with **70 automated JUnit 5 tests**.

EDUCATION

Bachelor of Computer Applications (BCA) — Shri Guru Ram Rai University, Dehradun **2025 – 2028**

Cumulative CGPA: 7.48 / 10.0 • **Semester 2 SGPA:** 7.96 / 10.0

LEADERSHIP & ACHIEVEMENTS

- **Secretary — TechMantra Coding Club:** Promoted from Core Member to Secretary within 10 months; organized 5+ hands-on coding bootcamps and technical workshops for 120+ student developers.
- **Startup Innovation Recognition:** Awarded Certificate of Recognition from **UCOST & SGRR University IIC** for pitching the startup concept *College Clash* in university innovation competitions.
- **Smart India Hackathon (SIH'26 Internal Round):** Awarded Certificate of Participation, collaborating in a collegiate software team to architect solutions for national-level problem statements.
- **Hackathon Competitions:** Awarded Certificate of Participation in **Algothon'26 (Algoxilla)**; engineered and deployed DevTinder backend in 72 hours during **Devlynix Buildathon 2.0**.
- **Algorithmic Problem Solving:** Solved data structures and algorithms problems in Java on LeetCode (@Nikunjgarg12), focusing on arrays, strings, two-pointers, and linked lists.